

Supplementary Information: IRSpectra-Bench — protocol, scale tables, cases, and reproducibility

Anonymous authors

Companion to the ICLR 2027 submission

Generation Recall, Not Verification, Binds LLM Structure Elucidation
from Literature Spectra

20 September 2026

This SI holds volume that does not belong on the 9-page ICLR main text. Every accuracy number is copied from a committed lock (docs/HEADLINE_N500_2026-09-20.md, spectro-agent PR #41 48fed30; docs/NIGHT_POOL_2026-09-20.md). No CIs are invented for $n=500$. The paper headline is $n=500$ (227/500 top-1; 249/500 recall). The main wall is the generation decomposition 227/22/251. Do not quote 243/519. Roster: frozen qid lists in the code release. Figure and listing numbers below refer to the main PDF (Fig. 1 = protocol + $n=500$ generation wall; Listing 1 = IRexp payload). They are not re-numbered here.

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1 How to read this SI (figure / listing map)

The night main-text plate and listing are the source of truth. This SI does not fight that numbering.

Table 1: Main-text floats vs this SI. Fig. 1 is the Figma Bro lead plate.

main label	object	SI role
Fig. fig1	Fig. 1 (fig1_lead_overview)	Lead: input $\rightarrow$ generate $\rightarrow$ verify, factorisation, and $n=500$ generation wall 227/22/251.
Fig. fig-chemspace	Fig. 2	Literature-cohort chemistry.
Listing lst:payload	Listing 1 (listing1_irexp_record)	IReXP JSON+molecule (NTJIYHYWVZY-BEX; $C_{12}H_{16}N_2O_2$). SMILES/InChIKey are resource fields, not solver payload.
Fig. fig-wall	appendix (fig_wall_diagnostic)	Repeat of the Fig. 1 $n=500$ generation wall.
Fig. fig-robustness	Fig. 4	Formula-only / recency / four-vendor.
Fig. fig-mechanism	Fig. 5	Locked v3-R25 verify-save. Reused in §9.
Fig. fig3-method	Fig. 6	Inference ladder on the 60-compound arm.
App. cases	worked modes	Same integers as §9.
SI Figs. S1–S3	reprints	Same three files as main Fig. 1 / Listing 1 / $n=500$ wall (§2). Not a second lead figure.

Equations in the main text are Eqs. (1)–(7): factorisation, InChIKey-14 indicator, top-1, recall@ k , prec./recall, ^{13}C chamfer, pooled estimator. They are restated in §4 with the same meaning. No new metric is defined.

2 Visual plates (reprints; not a second Figure 1)

The files below are the same as the main PDF. They are reprinted so this SI is readable alone. Main-text Figure 1 (fig1_lead_overview) remains the only lead diagram. Panel letters (a)/(b)/(c) from an earlier draft do not apply: the Figma Bro plate is one figure (protocol strip, factorisation, wall).

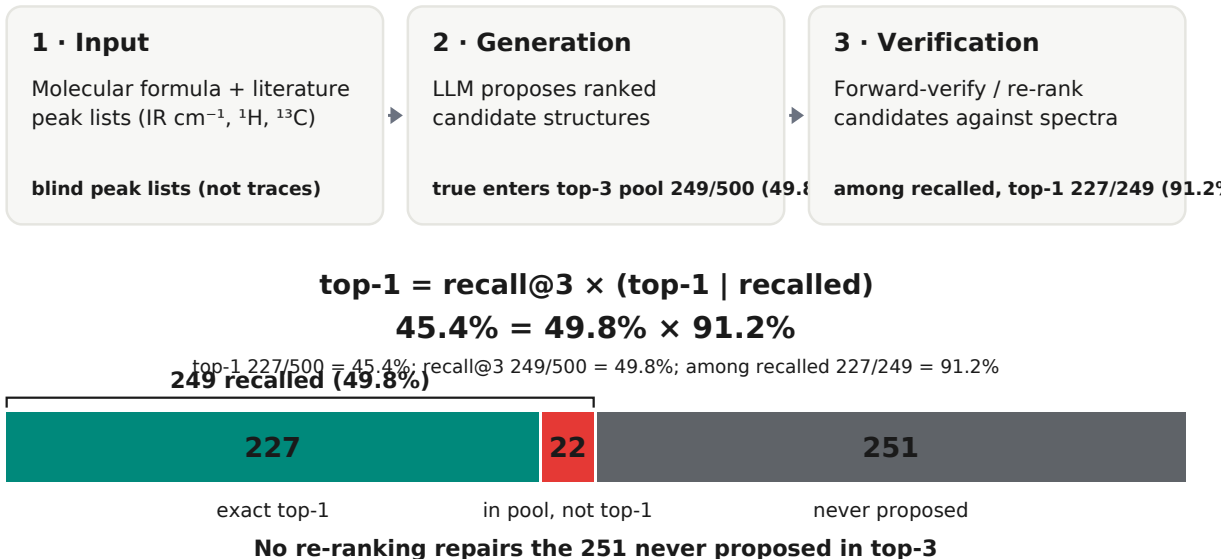


Figure S1: Reprint of main-text Figure 1. Input $\rightarrow$ generate $\rightarrow$ verify; $45.4\% = 49.8\% \times 91.2\%$; $n=500$ generation wall 227/22/251. Propose is the wall, not verify.

3 Blind protocol and pre-registration

3.1 Solver-facing contract

Each IRSpectra-Bench item is a four-field JSON object. The solver sees only

$\{\text{qid}, \text{formula}, \text{ir_bands_cm-1}, \text{h_nmr}, \text{c_nmr}\}$.

No name, SMILES, InChIKey, scaffold, atom-type hint, retrieved spectrum, licence stamp, or PMC id enters the prompt (main-text Listing 1, adapted from one IReXP record). Gold constitution is scored offline by the InChIKey-14 indicator (§4).

Difficulty is declared a priori by RDKit ring analysis: simple iff ≤ 2 rings, no fused/spiro/bridge-head system, ≤ 22 heavy atoms; else complex. The $n=500$ headline is near-50/50 (248/252); the eligible IReXP corpus is 17.5% simple / 82.5% complex. Reweighting lives on a validate-clean subset (§5) and is not an $n=500$ figure.

Listing 1. Example IRexp JSON record (commercial DoR).

```
{
  "id": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "ir_source": "experimental",
  "source_doi": "PMC:13029360",
  "pmcid": "PMC13029360",
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), 6.78 (d,
    J = 7.3 Hz, 2H, Ar), 6.25 (d, J = 8.1 Hz, 1H, NH), 4.10
    (hex, J = 8.1 Hz, 1H, CH), 3.66 (s, 3H, OCH3), 2.23-2.09
    (m, 2H, CH2), 1.90-1.71 (m, 2H, CH2), 1.64-1.51 (m, 2H,
    CH2)",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8",
  "smiles": "C0c1ccc(NC(=O)NC2CCC2)cc1",
  "inchikey": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "has_structure": true,
  "license": "CC-BY",
  "license_pool": "commercial"
}
```

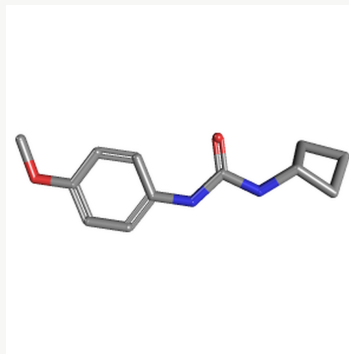


Figure S2: Reprint of main-text Listing 1 (listing1_irexp_record). IRexp JSON+molecule ($C_{12}H_{16}N_2O_2$; NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey are resource fields, not solver payload. Not a Data Descriptor dump. Locked v3-R25 is the mechanism case (Figure S4), not this listing.

3.2 Pre-registration pointers

These files live in the companion spectro-agent tree, not this postcard manuscript repo. Pointers are recorded so a reviewer can ask for the committed note rather than a reconstructed story.

- $n=500$ roster: frozen qid lists in the code release (do not reconstruct a composition story in prose). Sampler seeds for early draws were not captured (gap #4 in docs/SUBMISSION.md); the released roster is fixed and reusable but not exactly re-drawable.
- expand-500 campaign: docs/EXPANSION_PREREGISTRATION_500.md (seed 2026500; 115/115 simple/complex). 0 collisions with any prior round. Six ^{13}C -overread flags known before any solve: R26, R31, R102, R105, R107, R138.

3.3 Stopping / pooling rules

- The $n=500$ roster is the pre-specified frozen list (docs/headline500_expand200_qids.json plus the committed deposits), not an outcome-selected subset.
- score_pooled.py -expand-500 refuses a missing predictions2.jsonl or a withheld key.
- answers2.jsonl is not in either working tree (git ls-files empty for the withheld expansion keys).
- Keys were restored only under /tmp/blind/_key/ for the 2026-09-20 score pass, then deleted.
- validate_benchmark.py was not re-run on the night pass (would rewrite committed clean_qids.json). The existing 224-qid snapshot is used.
- collect_round.py was not run; 230 raw/single_R*.jsonl files were left untouched.
- forward_verify_main.py score did run on expand-500 (§8): 103/230 verify vs 129/230 self. Official prep was not re-run.
- -ci / bootstrap was not run on expand-500. Do not invent CIs.

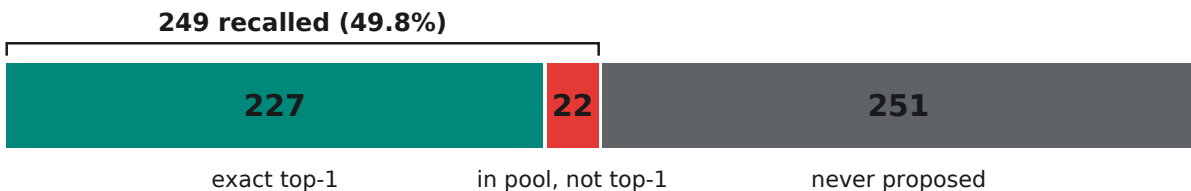


Figure S3: Reprint of the $n=500$ generation wall (fig_wall_diagnostic): 227 exact top-1 / 22 in pool, not top-1 / 251 never proposed. Same integers as Figure S1. Propose is the wall, not verify.

3.4 What the solver may do

RDKit formula / parse check only. No web search. No retrieved spectrum. No thinking tier on the core protocol. A thinking-tier consumer arm is labelled as such in the score notes and is not interchangeable with the no-thinking protocol.

A deposit is JSONL of {qid, candidates: [SMILES,...]} under a stated budget. Headline budget: ≤ 3 ranked SMILES per qid. Self-ranking vs forward-verified re-ranking must be labelled; they are not interchangeable.

4 Scoring contract (equations used in the main text)

These are the same identities as main-text Eqs. (1)–(7). They are restated so the SI is readable alone. No new metric is introduced.

A candidate $\hat{y}$ matches gold y^* at constitution iff

$$s(\hat{y}, y^*) = \mathbf{1}[\text{IK}_{14}(\hat{y}) = \text{IK}_{14}(y^*)]. \quad (1)$$

IK_{14} is the first 14 characters of the RDKit InChIKey (connectivity layer). Full stereo-sensitive InChIKey is reported only on validate-clean $n=295$ (§5).

With ranked pool $\mathcal{C}_i = (c_{i1}, \dots, c_{ik})$, $k \leq 3$,

$$\text{top-1} = \frac{1}{n} \sum_i s(c_{i1}, y_i^*), \quad (2)$$

$$\text{recall@}k = \frac{1}{n} \sum_i \mathbf{1}[\exists j \leq k : s(c_{ij}, y_i^*) = 1], \quad (3)$$

$$\text{prec.}|\text{recall} = \frac{\sum_i s(c_{i1}, y_i^*)}{\sum_i \mathbf{1}[y_i^* \in \mathcal{C}_i]}. \quad (4)$$

These are the three reportable numbers. They factorise as

$$\text{top-1} = (\text{recall@}k) \times (\text{prec.}|\text{recall}) \quad (5)$$

when the same pool is used for both stages — the rates still have different denominators and must not be differenced. Self-ranking prec.|recall on $n=500$ is 227/249 (91.2%). That is not forward-verify precision.

Forward-verify re-ranks by symmetric chamfer on ^{13}C peak sets $P, Q \subset \mathbb{R}$

$$d(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} |p - q| + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} |q - p|, \quad (6)$$

where $\hat{P}(c)$ is predicted blind to the observed list P_{obs} . Instrumented forward-verify diagnostic on a protocol slice (58/65; §8). expand-500 official score is in §8 (103/230 verify vs 129/230 self; no CIs; not a wall).

A pool estimator on an index set $\mathcal{S}$ is

$$\hat{p}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} s(c_{i1}, y_i^*). \quad (7)$$

The ICLR headline uses $\mathcal{S}$ of size 500 (frozen qid list). Do not quote the $n=519$ score_pooled.py print (243/519) as the headline.

Confidence intervals on earlier validate-clean subsets are bootstrap 95% (score_main.boot, seed 0, 2000 resamples). Paired model differences use McNemar with Holm correction. No CIs are reported for $n=500$ or expand-500.

5 Validate-clean extras (not the headline)

Source: docs/POOLED_HEADLINE_2026-09-16.md. These rows are not the $n=500$ headline. Self-ranking precision | recall on $n=500$ is 227/249 (91.2%) — not forward-verify precision.

¹³C-overread flags (R12, R22, R25, R82, R91) were marked before subset totals (scripts/validate_benchmark.py data/benchmark_expand/clean_qids.json).

Table 2: Validate-clean extras. Same scorer; not the headline.

metric		overall	simple	complex
scaffold-level (best Tanimoto ≥ 0.45)		64%	80%	48%
mean best Tanimoto		0.66	0.79	0.53
full InChIKey (stereo) top-1	93/295 (31.5%) [26–37]		—	—
full InChIKey recall	108/295 (36.6%) [31–42]		—	—
corpus-reweighted top-1	26.5% [21–32]		—	—
corpus-reweighted recall	31.3% [25–37]		—	—

Table 3: Size bins on validate-clean $n=295$.

heavy atoms	n	top-1	recall
≤ 15	53	36/53 (67.9%)	39/53 (73.6%)
16–25	152	65/152 (42.8%)	75/152 (49.3%)
> 25	90	15/90 (16.7%)	16/90 (17.8%)

On an instrumented slice, of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; only 22.6% share the true Murcko scaffold. Named-ring annotation on that slice is rare and does not drive the result.

6 expand-500 generation (NIGHT_POOL)

Source: spectro-agent data/benchmark_expand_500/NIGHT_POOL_2026-09-20.md on cursor/expand-bench-500-b78b (copied to docs/NIGHT_POOL_2026-09-20.md). Key restored only for score2, then deleted. answers2.jsonl is not in the tree.

Table 4: expand-500 generation (score2, InChIKey-14). No CIs.

set	<i>n</i>	top-1	recall@3
all deposited Opus	230	129/230 (56.1%)	138/230 (60.0%)
simple	115	82/115 (71%)	87/115 (76%)
complex	115	47/115 (41%)	51/115 (44%)
validate-clean	224	127/224 (56.7%)	135/224 (60.3%)
simple	110	81/110 (74%)	85/110 (77%)
complex	114	46/114 (40%)	50/114 (44%)

Mean best Tanimoto (score2 footer, all 230): simple 0.863; complex 0.645. Flagged qids (in 230, out of 224): R26, R31, R102, R105, R107, R138. Deposits: 230/230; predictions2.jsonl 230 lines, 690 candidates. Unique (formula, IR, ¹³C) vs irexp_resolved.jsonl.gz: 230/230; 0 prior-round InChIKey-14 collisions. Protocol: thinking-tier consumer arm — not interchangeable with the no-thinking core protocol.

Verbatim score2 footer (do not polish the integers):

overall recovered (top-3): 138/230 (60%) top-1 exact: 129/230 (56%)
 simple : recovered 87/115 (76%) top1 82/115 (71%) meanBestTani 0.863
 complex : recovered 51/115 (44%) top1 47/115 (41%) meanBestTani 0.645

7 Headline pool (generation only)

The estimator is Eq. 7. The paper headline is $n=500$. Source: docs/HEADLINE_N500_2026-09-20.md (spectro-agent 48fd30). Roster: frozen qid lists in the code release (docs/headline500_expand200_qids.jsonl)

Table 5: Headline generation. No CIs. No fverify. No composition column.

set	<i>n</i>	top-1	recall@3
paper headline	500	227/500 (45.4%)	249/500 (49.8%)

Strata on the $n=500$ headline (same InChIKey-14 rule): simple 161/248 (65%) top-1 / 172/248 (69%) recall; complex 66/252 (26%) / 77/252 (31%). Self-rank precision | recall: 227/249 (91.2%). Do not quote score_pooled.py -expand -expand-500 ($n=519$, 243/519). Do not draw a pooled forward-verify wall from these rows. Do not attach CIs.

8 Forward-verify coverage (honest gaps)

8.1 Instrumented protocol slice

Source: data/diagnosis.jsonl / scripts/forward_verify_all.py. This is a protocol-slice diagnostic, not the $n=500$ wall.

Main-text Figure 1 is the $n=500$ generation wall, not this diagnostic.

8.2 Other expansion fverify — not a pooled wall

No additional expansion forward-verify bundle is claimed as a wall. Self-ranking 227/249 on $n=500$ is not a substitute.

Table 6: Instrumented forward-verify diagnostic on a protocol slice. Not the $n=500$ generation wall.

	60-compound arm	protocol slice
generation recall	19/60 (32%)	65 (34%)
top-1, solver self-ranking	14/60 (23%)	55 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58 (30%)
precision recall — fverify	16/19 (84%)	58/65 (89%)
multi-candidate only — fverify	10/13 (77%)	30/37 (81%)
derangement floor on $n=37$	—	54.0%; fverify +27.1 points
McNemar vs self-rank	—	$p=0.55$ (7 gained, 4 lost)

8.3 expand-500 — official score (not a wall)

Source: spectro-agent data/fverify_expand_500/STATUS.md, results.txt, and NIGHT_POOL_2026-09-20.md (cursor/expand-bench-500-b78b). Blind ^{13}C campaign over 681 unique candidate SMILES (f1–f41 all present). Official forward_verify_main.py score() ran 2026-09-20 (is_true filled under /tmp only; committed candidates.jsonl still omits it; official prep was not re-run). No CIs. Verify does not beat self-rank. Headline is $n=500$ generation. This is not a pooled wall and does not replace the protocol-slice diagnosis.json diagnostic.

Table 7: expand-500 official chamfer score. Copied from NIGHT_POOL / fverify STATUS. Not a wall. Not the headline.

set	n	recall	self top-1	verify top-1	verify recall
all deposited	230	138/230 (60.0%)	129/230 (56.1%)	103/230 (44.8%)	103/138 (74.6%)
validate-clean	224	135/224 (60.3%)	127/224 (56.7%)	103/224 (46.0%)	103/135 (76.3%)

Coverage: 41/41 batches; 681/681 unique SMILES; 230/230 qids fully covered. Multi-candidate only: all-230 self 128/137, verify 102/137; clean-224 self 126/134, verify 102/134. forward_verify_all.py was not run (no expand-500 diagnosis.json).

HOSE-style lookup on the protocol-slice candidate sets ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%) — directional, not a new headline. Deranging observed ^{13}C pairings collapses precision ($p=0.001$).

9 Failure and success case panels

The three locked-slice modes in the main appendix, plus extra panels that use only already-reported numbers. IDs below are locked-round labels (expansion qids reuse R22/R25). No SMILES are invented for R06/R26; only v3-R25 molecules are drawn (Figure S4).

9.1 Proposal miss — main-R06 / $\text{C}_{29}\text{H}_{30}\text{FNO}_2$

True acridinedione never enters the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. This is a never-proposed miss: re-ranking cannot recover a structure that was never proposed (Eq. 5).

Table 8: Three-mode case grid (disclosed facts only). Same integers as main-text Table tab:cases.

mode	id / formula	disclosed facts
Proposal miss	main-R06 / $C_{29}H_{30}FNO_2$	True acridinedione never in the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. This is a never-proposed miss.
Verify-save	v3-R25 / $C_{10}H_{14}N_2O$	True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm). IR 1679 cm^{-1} . Figure S4. Not Listing 1.
Verify-fail	v3-R26 / $C_{14}H_{11}N_3O_3$	True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} .

9.2 Verify-save — v3-R25 / $C_{10}H_{14}N_2O$

True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm, Eq. 6). IR 1679 cm^{-1} . Main-text Figure 5; SI Figure S4.

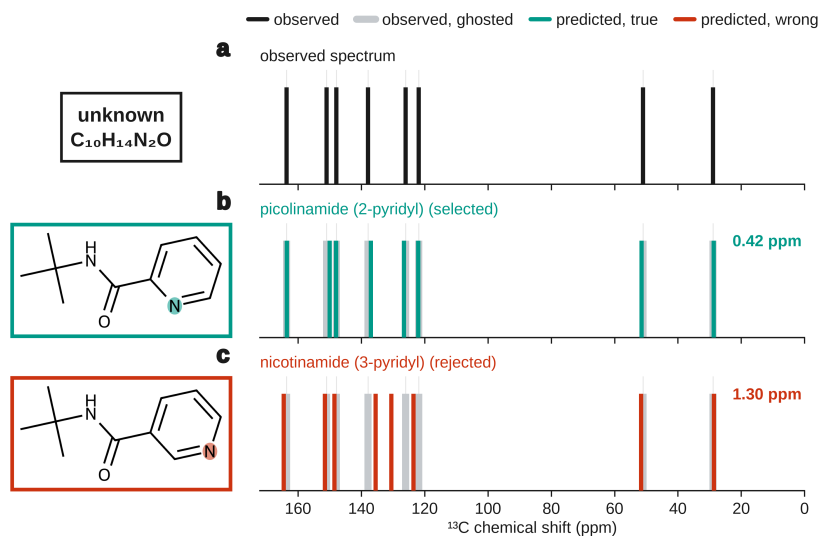


Figure S4: Locked v3-R25 verify-save (main-text Figure 5). Self-rank prefers 3-pyridyl; forward ^{13}C recovers 2-pyridyl (chamfer 0.42 vs 1.30 ppm). A verify-save, not a recall win. Not Listing 1.

9.3 Verify-fail — v3-R26 / $C_{14}H_{11}N_3O_3$

True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} . Ranking can fail when predicted ^{13}C lists are nearly tied. This is why we do not claim an accuracy advance from forward-verify (McNemar $p=0.55$; 7 gained, 4 lost on the protocol slice).

9.4 Listing 1 is the IReXP record plate

Main-text Listing 1 is listing1_irexp_record (C₁₂H₁₆N₂O₂ urea; InChIKey NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey / licence fields are resource metadata. Bench solvers see only formula and printed peak lists. Gold constitution is scored only by Eq. 1. Locked v3-R25 remains the mechanism worked case (main-text Figure 5), not Listing 1.

The plate is reprinted as Figure S2.

9.5 Extra panels (already-reported numbers)

Formula-only collapse (control, $n=60$). Formula only: 3/60 (5%) top-1 and 3/60 (5%) recall. Formula + IR + ¹H + ¹³C: 14/60 (23%) / 19/60 (32%). Outcomes perfectly nested (McNemar $p=0.001$). The model is reading the peak lists, not only the formula.

Generate-wide (60-compound arm). Ten solver agents, up to six candidates each, pooled then re-ranked: recall 32% → **42%**, top-1 23% → **30%** (McNemar $p=0.34$); verification precision falls 84% → 72%. The training-free ceiling remains recall-limited. This is a success-for-recall panel, not an accuracy claim.

IReXP-fine-tuned generator (locked slice). Pooled with Claude: recall 33.5% → 54.1%, top-1 28.4% → 35.1% (McNemar $p=0.015$). The wall is a property of training-free elicitation, not of the task. Training is the one intervention already seen to move $P(\text{true} \in \text{pool})$.

Battery-electrolyte subset. $n=46$ literature electrolyte chemistry (not operando): 26% top-1, recall-bound — same regime as the locked slice.

Four-model Claude ladder ($n=24$). Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable is capability-sensitive but underpowered for adjacent ranks, with a disclosed protocol asymmetry. Not a headline.

Difficulty variation. Pre-registered draws vary in difficulty. That is a result. The headline remains $n=500$ (Eq. 7).

10 Prompt skeletons

10.1 Headline solver-facing skeleton

Wrapping instruction text was not captured on the headline Claude arm (Limitations (i); docs/-SUBMISSION.md gap #1). Only the per-compound four-field payload is released. The skeleton below is the contract, not a recovered verbatim system prompt.

You are given only:

- formula (as from HRMS)
- ir_bands_cm-1 (author-transcribed experimental list)
- h_nmr, c_nmr (printed shift lists)

Return ≤ 3 ranked SMILES.

No name, no SMILES hint, no scaffold, no retrieved spectrum.

Check molecular formula with RDKit before emitting a candidate.

Listing 1 payload (fields only; ¹H abbreviated with ellipsis in the main float; gold held out):

```
{
  "formula": "C12H16N2O2",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), ...",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8"
}
```

10.2 Forward-verify skeleton

For each candidate SMILES, predict a ^{13}C peak list without seeing the observed list. Re-rank by $d(\hat{P}(c), P_{\text{obs}})$ (Eq. 6). The instruction dispatched to the forward-prediction sub-agents was likewise not captured; only the candidate SMILES and the resulting numeric lists are on disk for the protocol slice.

10.3 Cross-vendor harness

The verbatim prompts in the (unreleased-here) ESI S2 of the spectro-agent tree are the cross-vendor harness prompts, not the missing headline Claude wrapper. External submissions must document model version, prompt, and tool access (docs/LEADERBOARD.md).

10.4 Deposit schema

```
{"qid": "R01", "candidates": ["SMILES_rank1", "SMILES_rank2", "SMILES_rank3"]}
```

Score locally (spectro-agent):

```
python scripts/score_submission.py --predictions my_run.jsonl --name "YourModel-1.0"
python scripts/score_pooled.py --stereo # validate-clean stereo extras
python scripts/forward_verify_all.py # protocol-slice diagnostic
```

11 Hyperparameters, hardware, and model cards

11.1 Headline model card

11.2 Cross-vendor card ($n=60$)

Verification precision exceeds generation recall in every arm. Bootstrap separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$). Reasoning-effort tier per vendor arm was not recorded (gap #2).

11.3 Literature decomposition (copied, not re-derived)

Any system reporting top-1 = a and top- k = b implies recall $\geq b$ and conditional precision $\leq a/b$. Script: scripts/literature_decomposition.py.

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. The solver-alone precision column on $n=500$ is self-ranking (227/249); the forward-verify row is the instrumented protocol slice (58/65).

Table 9: Headline solver. Gaps are recorded, not filled.

item	value
headline solver	Claude Opus, consumer subscription
thinking tier (core protocol)	none; a labelled thinking-tier arm exists
snapshot / seed / temperature / top_p / top_k / max_tokens	not exposed; not recorded (SUBMISSION gaps #2-#3)
candidate budget	≤ 3 SMILES / qid
tool access	RDKit formula / parse check only; no web search
fine-tuning	none on the headline arm
paid API	no, for the core protocol
hardware	consumer harness; no cluster job for inference
scoring hardware	commodity CPU; RDKit InChIKey-14
cross-vendor arm	$n=60$; Grok 4.6, Gemini 3.7 Flash, GPT-5.6 Sol
Claude mean candidates on that arm	2.20 (others 3.00) — recall rankings approximate
expand-500 solver	Claude Opus thinking-tier (declared; not headline)
Fable incomplete arm	deposits not scored; not pooled
scoring scripts	scripts/score_pooled.py, score_main.py, bench-mark_v2.py score2

Table 10: Cross-vendor decomposition. Recall and precision use different denominators.

model	generation recall	prec. recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

12 Two-paper split and anonymity

IRSpectra-Bench (this ICLR paper) is the blind peak-list benchmark and the propose $\gg$ verify diagnosis. IRExp is a companion Scientific Data Data Descriptor (in preparation): 121,233 records; 43,060 structure-linked; 33,201 full IR+ ^1H + ^{13}C +structure quadruples. This SI cites that resource and does not re-present Methods / Data Records / licence-pool narrative.

Double-blind review copy: <https://anonymous.4open.science/r/peaklist-corpus-review-10C4/>. Named Hugging Face / GitHub hosting is restored at camera-ready. Do not put ilkhamfy/* URLs in the PDF. Fine-tunes must use data/train_no_bench.jsonl.gz so benchmark InChIKeys are withheld.

Listing 1 is the only IRExp-style JSON+molecule plate in the ICLR PDF. It is a pointer, not a second paper.

13 Reproducibility checklist

ICLR-style checklist, filled honestly. “N/A” is used only when the item does not apply; gaps are named.

1. Code / data / frozen deposits. Frozen questions / predictions live in spectro-agent. This postcard repo is the ICLR manuscript. Camera-ready will name the archival host. Roster: frozen qid lists in the code release.

Table 11: Selected literature decomposition rows (as in main-text Table 6).

system (n)	top-1	top- k (k)	recall	prec.
this work, solver alone (500)	45.4%	49.8% (3)	49.8%	91.2%
this work + forward-verify (slice)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver, literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent, literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales, education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales, industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, 6–13 atoms, NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, scaffold split, simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

2. Headline regeneration. Committed score notes regenerate 227/500 and 249/500 (docs/HEADLINE_N500_2026-09-20.md).
3. Protocol-slice fverify. python scripts/forward_verify_all.py regenerates the protocol-slice diagnostic (not the $n=500$ wall).
4. expand-500 score2. Restore /tmp/blind/_key/benchmark_expand_500.answers2.jsonl.withheld, then benchmark_v2.py score2 -outdir data/benchmark_expand_500. Re-withhold the key. Confirm git ls-files data/benchmark_expand_500/answers2.jsonl is empty.
5. Do not commit withheld answers2.jsonl keys.
6. Do not invent CIs for expand-500; -ci was not run.
7. expand-500 fverify score is already in NIGHT_POOL / fverify STATUS (103/230 verify vs 129/230 self). Do not invent CIs or a wall. Do not re-run official prep.
8. Double-blind. Companion bib is Anonymous; dataset URL in the PDF is only the anonymous.4open review copy.
9. Fine-tunes must use data/train_no_bench.jsonl.gz.
10. IReXP construction, licences, and transcription audit belong in the Scientific Data companion — not in this SI beyond Listing 1.
11. Inference is not bit-exact. No pinned snapshot, temperature, or seed (Limitations (i)). Distributional agreement of frozen deposits is the intended claim.
12. Uncaptured artefacts (cannot be recovered): headline Claude wrapper text; dated model snapshots; decoding parameters; main-round sampler seeds; closed-book audit transcripts (docs/SUBMISSION.md gaps #1–#5).
13. Deferred arms. Expert-chemist audit not run. Leave-one-modality-out abandoned. Parser extraction-recall audit deferred to Sci. Data. SpectraLLM / IR-Agent / Spectro / NMIRacle / Alberts / CASE not scored on this bench.
14. Figure rebuild limits. fig_wall_diagnostic / Fig. 1 wall is the $n=500$ generation decomposition (227/22/251). expand-500 has an official score() but no diagnosis.json, so a pooled forward-verify wall cannot be rebuilt.
15. This manuscript compile. python3 scripts/build_pdf.py (main); python3 scripts/build_si.py (this SI). Engine: tectonic 0.17.0 or pdfLaTeX+BibTeX / XeLaTeX+BibTeX.

13.1 What a reviewer can re-check without the withheld keys

- Every a/b ($c\%$) in the main text and this SI is internally consistent with the integers in docs/HEADLINE_N500_2026-09-20.md, docs/POOLED_HEADLINE_2026-09-16.md and docs/NIGHT_POOLED_2026-09-20.md.
- Fig. 1 caption, Limitations (vi), and Appendix FAQ agree: wall = $n=500$ generation (227/22/251; 227/500, 249/500).
- expand-500 all-230 rows carry “no CIs / thinking-tier / not a wall”. Official fverify is 103/230

verify vs 129/230 self (SI only; not a wall). Do not quote 243/519.

14 Reviewer FAQ (SI copy)

These answers restate main-text Limitations and Appendix B. They add no new metrics.

Why is top-1 45% if the wall figure looks worse? It should not. Figure 1 is the $n=500$ generation wall (227/22/251; 249/500 recalled). Headline is 227/500 (45.4%) and 249/500 (49.8%).

Is the roster a hidden mix? The $n=500$ roster is the frozen qid list in the code release. Do not reconstruct a composition story from protocol notes.

Where is the instrumented forward-verify wall? SI only, as a protocol-slice diagnostic (not the $n=500$ wall).

Where is expand-500 official fverify? This SI: 103/230 verify vs 129/230 self (§8); no CIs; not a wall. Headline is $n=500$ (Eq. 7).

Consumer harness / no snapshot / no thinking tier? Correct and intentional for the scored headline. Scoring of frozen deposits is reproducible; inference is not bit-exact.

Why no SpectraLLM / IR-Agent / CASE SOTA table? Different objects. None is scored on this bench. The paper is a stage diagnosis, not a method race.

Is IRexp a second ICLR paper? No. IRexp is a companion resource pointer (Listing 1; §12).

Should the headline move off $n=500$ or quote 243/519? No. Eq. 7 uses $\mathcal{S}$ of 500. Do not quote 243/519.

What this SI is not

A Data Descriptor. A pooled forward-verify wall. An expand-500 wall. A recommendation to switch the ICLR headline off $n=500$ or to quote 243/519. A source of invented CIs or new metrics.